



Koneru Lakshmaiah Education Foundation

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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(24CI3201)

A.Y. 2026 - 2027

TITLE OF THE PROJECT	Agile-Driven DevOps Hotel Management and Reservation Platform	
S. No.	University ID	Name
1	2420090052	A.Sharvani
2	2420030212	M.Vinusha
3	2420030170	Anjali Kasarla
4	2420030553	N.Phanindhra
Name of the Guide		RAJKUMAR PATIL

Abstract

StayWise: An AI-Powered Smart Hotel Management System is an intelligent web-based hotel management platform designed to automate and optimize hotel operations while enhancing the guest experience through Artificial Intelligence. The system integrates core hotel management functionalities such as room reservations, customer registration, room availability tracking, online payments, billing, housekeeping management, staff administration, and report generation into a centralized platform.

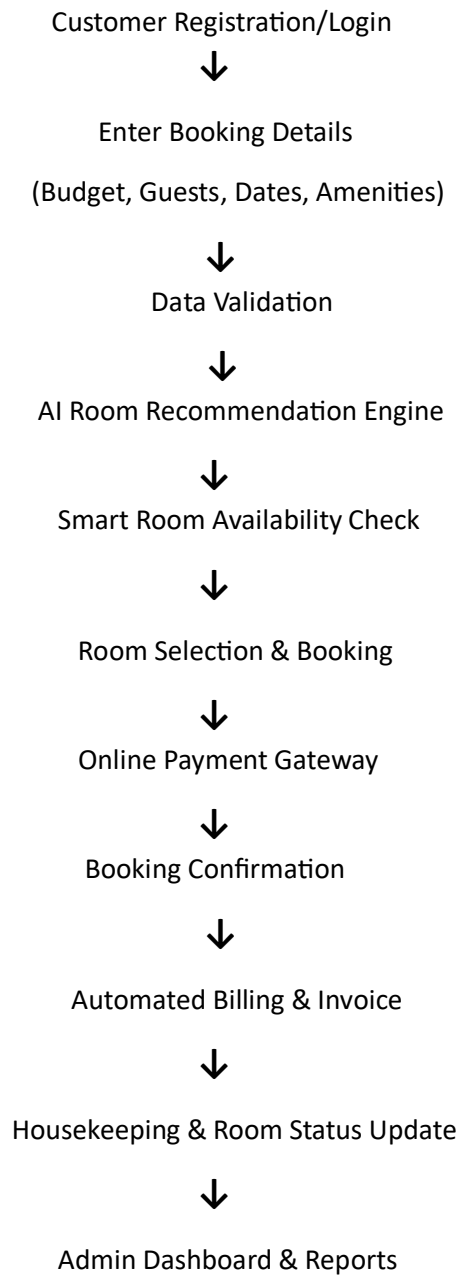
The project introduces advanced AI-driven features that distinguish it from traditional hotel management systems. The AI Room Recommendation Engine uses customer preferences, including budget, number of guests, travel purpose, room type, and desired amenities, to recommend the most suitable rooms using machine learning-based recommendation techniques. Additionally, the Smart Room Availability Calendar provides a real-time, color-coded visualization of room occupancy, reservations, check-ins, check-outs, maintenance status, and cleaning schedules, enabling efficient room allocation and operational planning.

The system follows the Agile Software Development Methodology, supporting iterative development, continuous integration, testing, and regular stakeholder feedback to ensure adaptability, reliability, and software quality. The application is developed using Java (Spring Boot) for backend services, HTML5, CSS3, JavaScript, Bootstrap, and Thymeleaf for the frontend, and MySQL as the relational database.

To improve recommendation accuracy, the system utilizes a hotel booking dataset containing room details, customer booking history, pricing, occupancy, amenities, seasonal demand, customer ratings, and booking preferences. The dataset is preprocessed through data cleaning, feature engineering, normalization, and encoding before training the recommendation model. Machine learning algorithms such as Random Forest, Decision Tree, or Content-Based Filtering can be employed to generate personalized room recommendations based on customer requirements.

The proposed system is evaluated based on functionality, recommendation accuracy, performance, security, usability, scalability, and reliability. Experimental results demonstrate that StayWise significantly reduces manual effort, improves booking efficiency, optimizes room utilization, enhances operational management, and delivers a personalized customer experience.

WorkFlow:



Abstract

Signature of the Guide

HOD